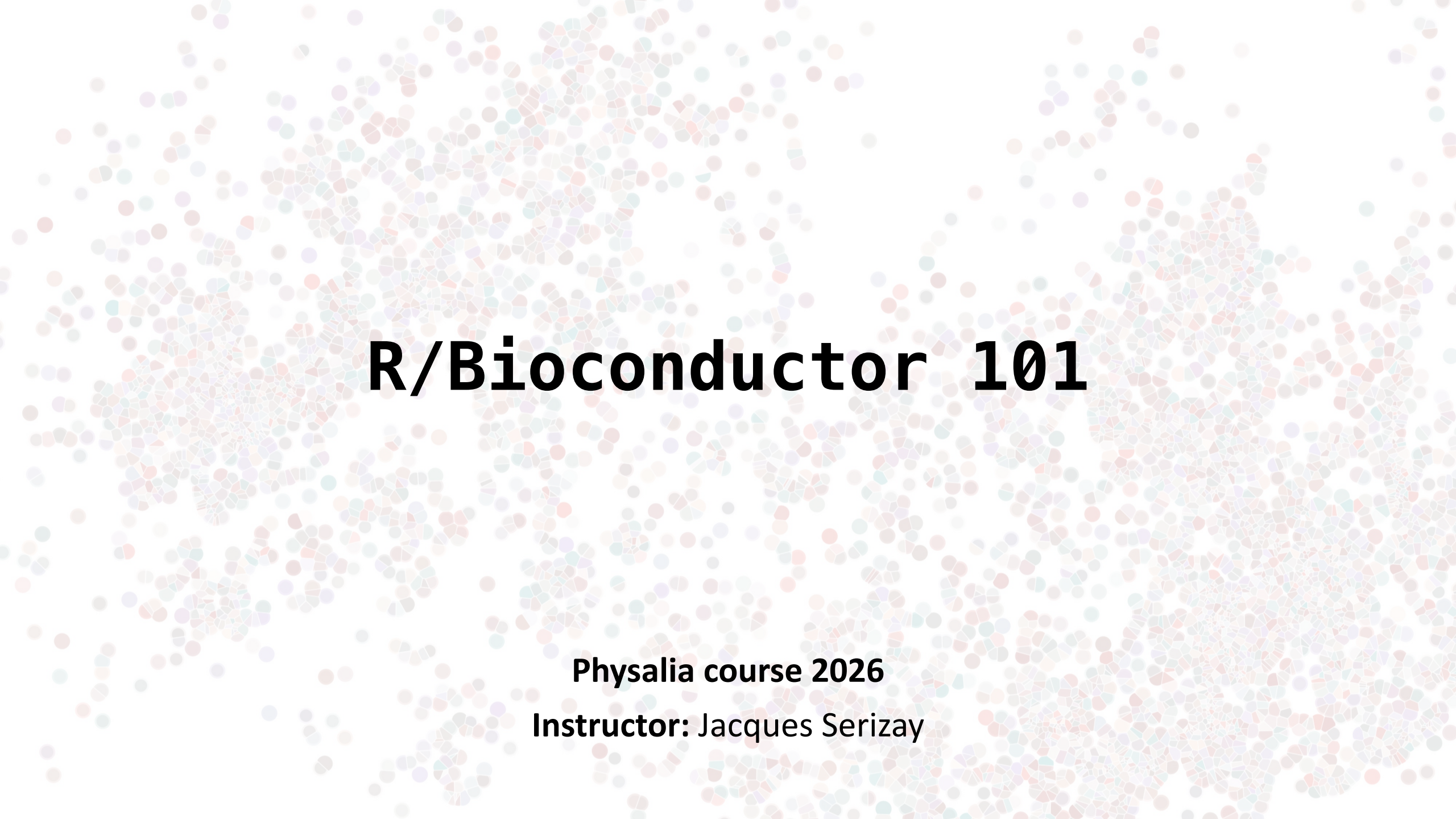




R/Bioconductor 101

Physalia course 2026

Instructor: Jacques Serizay

Vectors

- Defined with `c()` function
- All the elements must be from the same class
- Can be subset with `[...]`

```
r$> c(1, 2, 3)
[1] 1 2 3
```

```
r$> c('a', 'b', 'c')
[1] "a" "b" "c"
```

```
r$> c('a', 'b', 'c', 1, 2, 3)
[1] "a" "b" "c" "1" "2" "3"
```

```
r$> vec <- c(1, 2, 3)
```

```
r$> vec[2]
[1] 2
```

```
r$> vec[3]
[1] 3
```

```
r$> vec[4]
[1] NA
```

Tibbles

- Modern data.frame, tabular shape
- Created with tibble()

```
r$> library(tibble)
```

```
r$> tbl <- tibble(  
  "vec1" = c(4, 1, 2, 4),  
  "vec2" = c('a', 'b', 'c', 'd')  
)
```

```
r$> tbl
```

```
# A tibble: 4 × 2
```

	vec1	vec2
	<dbl>	<chr>
1	4	a
2	1	b
3	2	c
4	4	d

Tibbles

- Modern data.frame, tabular shape
- Created with `tibble()`
- Subset with `[ ..., ...]`

```
r$> summary(tbl)
      vec1      vec2
Min.   :1.00   Length:4
1st Qu.:1.75   Class :character
Median :3.00   Mode  :character
Mean   :2.75
3rd Qu.:4.00
Max.   :4.00
```

```
r$> tbl[1, ]
# A tibble: 1 x 2
   vec1 vec2
  <dbl> <chr>
1     4  a
```

```
r$> tbl[, 2]
# A tibble: 4 x 1
   vec2
  <chr>
1  a
2  b
3  c
4  d
```

Tibbles

- Modern data.frame, tabular shape
- Created with `tibble()`
- Subset with `[ ..., ...]`
- Columns can also be accessed with `[ [...]]` or `$`

```
r$> tbl[1, 2]
# A tibble: 1 × 1
  vec2
  <chr>
1 a

r$> tbl$vec2
[1] "a" "b" "c" "d"

r$> tbl[['vec2']]
[1] "a" "b" "c" "d"
```

Tibbles

- Modern data.frame, tabular shape
- Created with `tibble()`
- Subset with `[ ..., ...]`
- Columns can also be accessed with `[...] ]` or `$`

```
tbl [ tbl$vec1 == 4 , "vec2" ]
```

```
r$> tbl[1, 2]
# A tibble: 1  1
  vec2
  <chr>
1 a
```

```
r$> tbl$vec2
[1] "a" "b" "c" "d"
```

```
r$> tbl[['vec2']]
[1] "a" "b" "c" "d"
```

```
r$> tbl [ tbl$vec1 == 4 , "vec2" ]
# A tibble: 2  1
  vec2
  <chr>
1 a
2 d
```

Lists

- Created with `list()` function
- Each element can be whatever object you want
- Each element can be named

```
r$> l <- list(
  first = LETTERS[1:3],
  second = NA,
  third = seq(10, 20),
  fourth = "bonjour",
  fifth = lm(Y ~ x, data = tibble(x = 1:5, Y = 4:8))
)

r$> l
$first
[1] "A" "B" "C"

$second
[1] NA

$third
[1] 10 11 12 13 14 15 16 17 18 19 20

$fourth
[1] "bonjour"

$fifth

Call:
lm(formula = Y ~ x, data = tibble(x = 1:5, Y = 4:8))

Coefficients:
(Intercept)          x
           3           1
```

Lists

- Created with `list()` function
- Each element can be whatever object you want
- Each element can be named
- Elements can be accessed using `$` or `[[...]]`

```
r$> l[[1]]
[1] "A" "B" "C"

r$> l[['first']]
[1] "A" "B" "C"

r$> l$first
[1] "A" "B" "C"

r$> summary(l)
      Length Class  Mode
first    3    -none- character
second   1    -none- logical
third   11    -none- numeric
fourth   1    -none- character
fifth   12     lm      list
```


Lists

- Created with `list()` function
- Each element can be whatever object you want
- Each element can be named
- Elements can be accessed using `$` or `[[...]]`
- One can “map” or “apply” (`lapply`) a function over each element of a list

```
r$> library(purrr)

r$> map(l, class)
$first
[1] "character"

$second
[1] "logical"

$third
[1] "integer"

$fourth
[1] "character"

$fifth
[1] "lm"

r$> map(l, ~ .x[3])
$first
[1] "C"

$second
[1] NA

$third
[1] 12

$fourth
[1] NA

$fifth
$fifth$effects
      (Intercept)              x
-1.341641e+01  3.162278e+00 -3.330669e-16  0.000000e+00  4.440892e-16
```

Tidyverse (<https://rstudio-education.github.io/tidyverse-cookbook/program.html>)

- Verb-based ecosystem
- `dplyr::filter`
- `dplyr::arrange`
- `dplyr::mutate`
- `purrr::map`
- `tidyr::pivot_*`
- `ggplot2` plotting functions

Everything documented here:

<https://www.r-bloggers.com/2020/12/the-tidyverse-in-a-table/>

Native `|>` pipe

- Just like a pipe in bash, for R
- Very useful in combination with tidyverse's dplyr for data wrangling

```
r$> glimpse(mtcars)
Rows: 32
Columns: 11
$ mpg <dbl> 21.0, 21.0, 22.8, 21.4, 18.7, 18.1, 14.3, 24.4, 22
$ cyl <dbl> 6, 6, 4, 6, 8, 6, 8, 4, 4, 6, 6, 8, 8, 8, 8, 8,
$ disp <dbl> 160.0, 160.0, 108.0, 258.0, 360.0, 225.0, 360.0, 1
$ hp <dbl> 110, 110, 93, 110, 175, 105, 245, 62, 95, 123, 123
$ drat <dbl> 3.90, 3.90, 3.85, 3.08, 3.15, 2.76, 3.21, 3.69, 3.
$ wt <dbl> 2.620, 2.875, 2.320, 3.215, 3.440, 3.460, 3.570, 3
$ qsec <dbl> 16.46, 17.02, 18.61, 19.44, 17.02, 20.22, 15.84, 2
$ vs <dbl> 0, 0, 1, 1, 0, 1, 0, 1, 1, 1, 1, 0, 0, 0, 0, 0, 0,
$ am <dbl> 1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
$ gear <dbl> 4, 4, 4, 3, 3, 3, 3, 4, 4, 4, 4, 3, 3, 3, 3, 3, 3,
$ carb <dbl> 4, 4, 1, 1, 2, 1, 4, 2, 2, 4, 4, 3, 3, 3, 4, 4, 4,
```

```
r$> mtcars |>
  as_tibble(rownames = 'car') |>
  separate(car, c('brand', 'type'), ' ', extra = 'merge') |>
  group_by(brand) |>
  mutate(over20mpg = mpg > 20) |>
  count(over20mpg) |>
  knitr::kable()
```

brand	over20mpg	n
:----- :----- :--		
AMC	FALSE	1
Cadillac	FALSE	1
Camaro	FALSE	1
Chrysler	FALSE	1
Datsun	TRUE	1
Dodge	FALSE	1
Duster	FALSE	1
Ferrari	FALSE	1
Fiat	TRUE	2
Ford	FALSE	1
Honda	TRUE	1
Hornet	FALSE	1
Hornet	TRUE	1
Lincoln	FALSE	1
Lotus	TRUE	1
Maserati	FALSE	1
Mazda	TRUE	2
Merc	FALSE	5
Merc	TRUE	2
Pontiac	FALSE	1
Porsche	TRUE	1
Toyota	TRUE	2
Valiant	FALSE	1
Volvo	TRUE	1

R *per se* is useful for statistical analyses.

Why do bioinformaticians keep talking about R then?

In other words, how do we unlock the power of R-stats in genomics?

What do you need in bioinformatics to study genomics?

Most common genomic files:

- BED format: essentially a set of chromosomal ranges
- BigWig format: essentially veeeeeeeeee...eeeeery long numerical vectors
- Fasta format: letters, letters, letters
- Others (bam, GFF, ...): can usually be described/built on as one of the two options above

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```
if (!require  
install.pack  
} BiocManage
```

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```
ly = TRUE))  
nction.")
```

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>

- As a package

```
r$> install.packages('BiocManager', repos='http://cran.us.r-project.org')
Installing package into '/home/rsg/R/x86_64-pc-linux-gnu-library/4.3'
(as 'lib' is unspecified)
trying URL 'http://cran.us.r-project.org/src/contrib/BiocManager_1.30.21.tar.gz'
Content type 'application/x-gzip' length 582625 bytes (568 KB)
=====
downloaded 568 KB

* installing *source* package 'BiocManager' ...
** package 'BiocManager' successfully unpacked and MD5 sums checked
** using staged installation
** R
** inst
** byte-compile and prepare package for lazy loading
** help
*** installing help indices
** building package indices
** installing vignettes
** testing if installed package can be loaded from temporary location
** testing if installed package can be loaded from final location
** testing if installed package keeps a record of temporary installation path
* DONE (BiocManager)

The downloaded source packages are in
      '/tmp/RtmptaV2XQ/downloaded_packages'
```

Bioconductor installation

- As a package
- Integrated in R
- Bioconductor's version depends on your R version
- Some Bioc packages are restricted to a certain version!

```
r$> install.packages('BiocManager', repos='http://cran.us.r-project.org')
Installing package into '/home/rsg/R/x86_64-pc-linux-gnu-library/4.3'
(as 'lib' is unspecified)
trying URL 'http://cran.us.r-project.org/src/contrib/BiocManager_1.30.21.tar.gz'
Content type 'application/x-gzip' length 582625 bytes (568 KB)
=====
downloaded 568 KB

* installing *source* package 'BiocManager' ...
** package 'BiocManager' successfully unpacked and MD5 sums checked
** using staged installation
** R
** inst
** byte-compile and prepare package for lazy loading
** help
*** installing help indices
** building package indices
** installing vignettes
** testing if installed package can be loaded from temporary location
** testing if installed package can be loaded from final location
** testing if installed package keeps a record of temporary installation path
* DONE (BiocManager)

The downloaded source packages are in
      '/tmp/RtmptaV2XQ/downloaded_packages'

r$> library(BiocManager)
Bioconductor version 3.17 (BiocManager 1.30.21), R 4.3.0 (2023-04-21)
```

Bioconductor packages

- Bioconductor packages are on Bioconductor, not CRAN
- So you install them using Bioconductor's BiocManager!

```
r$> install.packages('nullranges')
Installing package into '/home/rsg/R/x86_64-pc-linux-gnu-library/4.3'
(as 'lib' is unspecified)

Warning message:
package 'nullranges' is not available for this version of R

A version of this package for your version of R might be available elsewhere,
see the ideas at
https://cran.r-project.org/doc/manuals/r-patched/R-admin.html#Installing-packages

r$> BiocManager::install("nullranges")
Bioconductor version 3.17 (BiocManager 1.30.21), R 4.3.0 (2023-04-21)
Installing package(s) 'nullranges'
trying URL 'https://bioconductor.org/packages/3.17/bioc/src/contrib/nullranges_1.6.2.tar.gz'
Content type 'application/x-gzip' length 4935234 bytes (4.7 MB)
=====
downloaded 4.7 MB

* installing *source* package 'nullranges' ...
** using staged installation
** R
** inst
** byte-compile and prepare package for lazy loading
** help
*** installing help indices
*** copying figures
** building package indices
** installing vignettes
** testing if installed package can be loaded from temporary location
** testing if installed package can be loaded from final location
** testing if installed package keeps a record of temporary installation path
* DONE (nullranges)
```

Ep The downloaded source packages are in
'/tmp/RtmptaV2XQ/downloaded_packages'

- **GRanges** (through GenomicRanges package)
- **XNAStrings** (through Biostrings)
- **Import/export** from/to common genomic files (through BiocIO and rtracklayer packages)

- Workhorse class of Bioconductor
- Used to describe genomic intervals

```
> peaks <- rtracklayer::import('Share/day03/results/bwa/mergedLibrary/macsnarrowPeak/Reb1_R1_peaks.narrowPeak')
> peaks
GRanges object with 369 ranges and 6 metadata columns:
      seqnames      ranges strand |          name      score signalValue    pValue    qValue      peak
      <Rle>      <IRanges> <Rle> | <character> <numeric> <numeric> <numeric> <numeric> <integer>
[1]      I      102-492      * | Reb1_R1_peak_1       81      6.32656    10.6179    8.16077       28
[2]      I  92560-92807      * | Reb1_R1_peak_2      118      8.90459    14.3647   11.80700      105
[3]     II   5846-6092      * | Reb1_R1_peak_3       63      6.16472     8.7750    6.36486       51
[4]     II 111226-111389      * | Reb1_R1_peak_4     1714     63.70210   175.6310  171.48900       76
[5]     II 124859-125004      * | Reb1_R1_peak_5      397     20.54910    42.7364   39.77400       99
...     ...         ...     ... | ...         ...     ...     ...     ...     ...
[365]   XVI 840491-840698      * | Reb1_R1_peak_365      63      6.13093     8.80750    6.39684       98
[366]   XVI 844287-844438      * | Reb1_R1_peak_366      17      3.42484     3.93128    1.76562       88
[367]   XVI 870371-870586      * | Reb1_R1_peak_367     292     16.43930   32.08000   29.23800      161
[368]   XVI 899847-900090      * | Reb1_R1_peak_368    1279     45.43150  131.80100  127.92500      175
[369]   XVI 942584-942868      * | Reb1_R1_peak_369     162     10.95950   18.90550   16.25210      111
-----
seqinfo: 17 sequences from an unspecified genome; no seqlengths
```

- Workhorse class of Bioconductor
- Used to describe genomic intervals

seqnames(x) -> chromosome names

```
> peaks <- rtracklayer::import('Share/day03/results/bwa/mergedLibrary/macs/narrowPeak/Reb1_R1_peaks.narrowPeak')
> peaks
GRanges object with 369 ranges and 6 metadata columns:
```

	seqnames	ranges	strand	name	score	signalValue	pValue	qValue	peak
	<Rle>	<IRanges>	<Rle>	<character>	<numeric>	<numeric>	<numeric>	<numeric>	<integer>
[1]	I	102-492	*	Reb1_R1_peak_1	81	6.32656	10.6179	8.16077	28
[2]	I	92560-92807	*	Reb1_R1_peak_2	118	8.90459	14.3647	11.80700	105
[3]	II	5846-6092	*	Reb1_R1_peak_3	63	6.16472	8.7750	6.36486	51
[4]	II	111226-111389	*	Reb1_R1_peak_4	1714	63.70210	175.6310	171.48900	76
[5]	II	124859-125004	*	Reb1_R1_peak_5	397	20.54910	42.7364	39.77400	99
...	...	...	...	...	...	...	...	...	...
[365]	XVI	840491-840698	*	Reb1_R1_peak_365	63	6.13093	8.80750	6.39684	98
[366]	XVI	844287-844438	*	Reb1_R1_peak_366	17	3.42484	3.93128	1.76562	88
[367]	XVI	870371-870586	*	Reb1_R1_peak_367	292	16.43930	32.08000	29.23800	161
[368]	XVI	899847-900090	*	Reb1_R1_peak_368	1279	45.43150	131.80100	127.92500	175
[369]	XVI	942584-942868	*	Reb1_R1_peak_369	162	10.95950	18.90550	16.25210	111

```
-----
seqinfo: 17 sequences from an unspecified genome; no seqlengths
```

- Workhorse class of Bioconductor
- Used to describe genomic intervals
start(x) -> interval start

```
> peaks <- rtracklayer::import('Share/day03/results/bwa/mergedLibrary/macs/narrowPeak/Reb1_R1_peaks.narrowPeak')
> peaks
GRanges object with 369 ranges and 6 metadata columns:
      seqnames      ranges strand |      name      score signalValue    pValue    qValue      peak
      <Rle>      <IRanges> <Rle> | <character> <numeric> <numeric> <numeric> <numeric> <integer>
[1]      I      102-492      * | Reb1_R1_peak_1      81      6.32656    10.6179    8.16077      28
[2]      I    92560-92807      * | Reb1_R1_peak_2     118      8.90459    14.3647   11.80700     105
[3]     II     5846-6092      * | Reb1_R1_peak_3      63      6.16472     8.7750    6.36486      51
[4]     II 111226-111389      * | Reb1_R1_peak_4    1714     63.70210   175.6310  171.48900      76
[5]     II 124859-125004      * | Reb1_R1_peak_5     397     20.54910    42.7364   39.77400      99
...     ...      ...      ... | ...      ...      ...      ...      ...      ...
[365]    XVI 840491-840698      * | Reb1_R1_peak_365      63      6.13093     8.80750    6.39684      98
[366]    XVI 844287-844438      * | Reb1_R1_peak_366      17      3.42484     3.93128    1.76562      88
[367]    XVI 870371-870586      * | Reb1_R1_peak_367     292     16.43930   32.08000   29.23800     161
[368]    XVI 899847-900090      * | Reb1_R1_peak_368    1279     45.43150  131.80100  127.92500     175
[369]    XVI 942584-942868      * | Reb1_R1_peak_369     162     10.95950   18.90550   16.25210     111
-----
seqinfo: 17 sequences from an unspecified genome; no seqlengths
```


- Workhorse class of Bioconductor
- Used to describe genomic intervals

`end(x)` -> interval end

```
> peaks <- rtracklayer::import('Share/day03/results/bwa/mergedLibrary/macsnarrowPeak/Reb1_R1_peaks.narrowPeak')
> peaks
GRanges object with 369 ranges and 6 metadata columns:
```

	seqnames	ranges	strand	name	score	signalValue	pValue	qValue	peak
	<Rle>	<IRanges>	<Rle>	<character>	<numeric>	<numeric>	<numeric>	<numeric>	<integer>
[1]	I	102-492	*	Reb1_R1_peak_1	81	6.32656	10.6179	8.16077	28
[2]	I	92560-92807	*	Reb1_R1_peak_2	118	8.90459	14.3647	11.80700	105
[3]	II	5846-6092	*	Reb1_R1_peak_3	63	6.16472	8.7750	6.36486	51
[4]	II	111226-111389	*	Reb1_R1_peak_4	1714	63.70210	175.6310	171.48900	76
[5]	II	124859-125004	*	Reb1_R1_peak_5	397	20.54910	42.7364	39.77400	99
...	...	...	...	...	...	...	...	...	...
[365]	XVI	840491-840698	*	Reb1_R1_peak_365	63	6.13093	8.80750	6.39684	98
[366]	XVI	844287-844438	*	Reb1_R1_peak_366	17	3.42484	3.93128	1.76562	88
[367]	XVI	870371-870586	*	Reb1_R1_peak_367	292	16.43930	32.08000	29.23800	161
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[369]	XVI	942584-942868	*	Reb1_R1_peak_369	162	10.95950	18.90550	16.25210	111

```
-----
seqinfo: 17 sequences from an unspecified genome; no seqlengths
```


- Workhorse class of Bioconductor
- Used to describe genomic intervals

strand(x)

```
> peaks <- rtracklayer::import('Share/day03/results/bwa/mergedLibrary/macsnarrowPeak/Reb1_R1_peaks.narrowPeak')
> peaks
GRanges object with 369 ranges and 6 metadata columns:
```

	seqnames	ranges	strand	name	score	signalValue	pValue	qValue	peak
	<Rle>	<IRanges>	<Rle>	<character>	<numeric>	<numeric>	<numeric>	<numeric>	<integer>
[1]	I	102-492	*	Reb1_R1_peak_1	81	6.32656	10.6179	8.16077	28
[2]	I	92560-92807	*	Reb1_R1_peak_2	118	8.90459	14.3647	11.80700	105
[3]	II	5846-6092	*	Reb1_R1_peak_3	63	6.16472	8.7750	6.36486	51
[4]	II	111226-111389	*	Reb1_R1_peak_4	1714	63.70210	175.6310	171.48900	76
[5]	II	124859-125004	*	Reb1_R1_peak_5	397	20.54910	42.7364	39.77400	99
...	...	...	...	...	...	...	...	...	...
[365]	XVI	840491-840698	*	Reb1_R1_peak_365	63	6.13093	8.80750	6.39684	98
[366]	XVI	844287-844438	*	Reb1_R1_peak_366	17	3.42484	3.93128	1.76562	88
[367]	XVI	870371-870586	*	Reb1_R1_peak_367	292	16.43930	32.08000	29.23800	161
[368]	XVI	899847-900090	*	Reb1_R1_peak_368	1279	45.43150	131.80100	127.92500	175
[369]	XVI	942584-942868	*	Reb1_R1_peak_369	162	10.95950	18.90550	16.25210	111

```
-----
seqinfo: 17 sequences from an unspecified genome; no seqlengths
```

- Workhorse class of Bioconductor
- Used to describe genomic intervals

`mcols(x)` -> all metadata

```
> peaks <- rtracklayer::import('Share/day03/results/bwa/mergedLibrary/macsnarrowPeak/Reb1_R1_peaks.narrowPeak')
> peaks
GRanges object with 369 ranges and 6 metadata columns:
```

	seqnames	ranges	strand	name	score	signalValue	pValue	qValue	peak
	<Rle>	<IRanges>	<Rle>	<character>	<numeric>	<numeric>	<numeric>	<numeric>	<integer>
[1]	I	102-492	*	Reb1_R1_peak_1	81	6.32656	10.6179	8.16077	28
[2]	I	92560-92807	*	Reb1_R1_peak_2	118	8.90459	14.3647	11.80700	105
[3]	II	5846-6092	*	Reb1_R1_peak_3	63	6.16472	8.7750	6.36486	51
[4]	II	111226-111389	*	Reb1_R1_peak_4	1714	63.70210	175.6310	171.48900	76
[5]	II	124859-125004	*	Reb1_R1_peak_5	397	20.54910	42.7364	39.77400	99
...	...	...	...	...	...	...	...	...	...
[365]	XVI	840491-840698	*	Reb1_R1_peak_365	63	6.13093	8.80750	6.39684	98
[366]	XVI	844287-844438	*	Reb1_R1_peak_366	17	3.42484	3.93128	1.76562	88
[367]	XVI	870371-870586	*	Reb1_R1_peak_367	292	16.43930	32.08000	29.23800	161
[368]	XVI	899847-900090	*	Reb1_R1_peak_368	1279	45.43150	131.80100	127.92500	175
[369]	XVI	942584-942868	*	Reb1_R1_peak_369	162	10.95950	18.90550	16.25210	111

```
-----
seqinfo: 17 sequences from an unspecified genome; no seqlengths
```

- Workhorse class of Bioconductor
- Used to describe genomic intervals

`x$score`

`mcols(x)$score -> specific metadata`

```
> peaks <- rtracklayer::import('Share/day03/results/bwa/mergedLibrary/macsnarrowPeak/Reb1_R1_peaks.narrowPeak')
> peaks
GRanges object with 369 ranges and 6 metadata columns:
```

	seqnames	ranges	strand	name	score	signalValue	pValue	qValue	peak
	<Rle>	<IRanges>	<Rle>	<character>	<numeric>	<numeric>	<numeric>	<numeric>	<integer>
[1]	I	102-492	*	Reb1_R1_peak_1	81	6.32656	10.6179	8.16077	28
[2]	I	92560-92807	*	Reb1_R1_peak_2	118	8.90459	14.3647	11.80700	105
[3]	II	5846-6092	*	Reb1_R1_peak_3	63	6.16472	8.7750	6.36486	51
[4]	II	111226-111389	*	Reb1_R1_peak_4	1714	63.70210	175.6310	171.48900	76
[5]	II	124859-125004	*	Reb1_R1_peak_5	397	20.54910	42.7364	39.77400	99
...	...	...	...	...	...	...	...	...	...
[365]	XVI	840491-840698	*	Reb1_R1_peak_365	63	6.13093	8.80750	6.39684	98
[366]	XVI	844287-844438	*	Reb1_R1_peak_366	17	3.42484	3.93128	1.76562	88
[367]	XVI	870371-870586	*	Reb1_R1_peak_367	292	16.43930	32.08000	29.23800	161
[368]	XVI	899847-900090	*	Reb1_R1_peak_368	1279	45.43150	131.80100	127.92500	175
[369]	XVI	942584-942868	*	Reb1_R1_peak_369	162	10.95950	18.90550	16.25210	111

```
-----
seqinfo: 17 sequences from an unspecified genome; no seqlengths
```

- Workhorse class of Bioconductor
- Used to describe genomic intervals

```
> peaks <- rtracklayer::import('Share/day03/results/bwa/mergedLibrary/macsnarrowPeak/Reb1_R1_peaks.narrowPeak')
> peaks
GRanges object with 369 ranges and 6 metadata columns:
```

	seqnames	ranges	strand	name	score	signalValue	pValue	qValue	peak
	<Rle>	<IRanges>	<Rle>	<character>	<numeric>	<numeric>	<numeric>	<numeric>	<integer>
[1]	I	102-492	*	Reb1_R1_peak_1	81	6.32656	10.6179	8.16077	28
[2]	I	92560-92807	*	Reb1_R1_peak_2	118	8.90459	14.3647	11.80700	105
[3]	II	5846-6092	*	Reb1_R1_peak_3	63	6.16472	8.7750	6.36486	51
[4]	II	111226-111389	*	Reb1_R1_peak_4	1714	63.70210	175.6310	171.48900	76
[5]	II	124859-125004	*	Reb1_R1_peak_5	397	20.54910	42.7364	39.77400	99
...	...	...	...	...	...	...	...	...	...
[365]	XVI	840491-840698	*	Reb1_R1_peak_365	63	6.13093	8.80750	6.39684	98
[366]	XVI	844287-844438	*	Reb1_R1_peak_366	17	3.42484	3.93128	1.76562	88
[367]	XVI	870371-870586	*	Reb1_R1_peak_367	292	16.43930	32.08000	29.23800	161
[368]	XVI	899847-900090	*	Reb1_R1_peak_368	1279	45.43150	131.80100	127.92500	175
[369]	XVI	942584-942868	*	Reb1_R1_peak_369	162	10.95950	18.90550	16.25210	111

```
-----
seqinfo: 17 sequences from an unspecified genome; no seqlengths
```

seqinfo(x) -> genome information

Action functions

- ...[...] (to subset)
- shift()
- resize()
- reduce()
- coverage()
- ...

```
> peaks[2:6]
GRanges object with 5 ranges and 6 metadata columns:
      seqnames      ranges strand |      name      score signalValue      pValue      qValue      peak
      <Rle>      <IRanges> <Rle> | <character> <numeric> <numeric> <numeric> <numeric> <integer>
[1]   chrI    92560-92807      * | Reb1_R1_peak_2    118      8.90459    14.3647    11.80700      105
[2]   chrII     5846-6092      * | Reb1_R1_peak_3     63      6.16472     8.7750     6.36486       51
[3]   chrII  111226-111389      * | Reb1_R1_peak_4   1714     63.70210   175.6310   171.48900       76
[4]   chrII  124859-125004      * | Reb1_R1_peak_5    397     20.54910    42.7364    39.77400       99
[5]   chrII  135791-136046      * | Reb1_R1_peak_6    452     22.60400    48.2640    45.24710      132
-----
seqinfo: 17 sequences from an unspecified genome; no seqlengths
> shift(peaks[2:6])
GRanges object with 5 ranges and 6 metadata columns:
      seqnames      ranges strand |      name      score signalValue      pValue      qValue      peak
      <Rle>      <IRanges> <Rle> | <character> <numeric> <numeric> <numeric> <numeric> <integer>
[1]   chrI    92560-92807      * | Reb1_R1_peak_2    118      8.90459    14.3647    11.80700      105
[2]   chrII     5846-6092      * | Reb1_R1_peak_3     63      6.16472     8.7750     6.36486       51
[3]   chrII  111226-111389      * | Reb1_R1_peak_4   1714     63.70210   175.6310   171.48900       76
[4]   chrII  124859-125004      * | Reb1_R1_peak_5    397     20.54910    42.7364    39.77400       99
[5]   chrII  135791-136046      * | Reb1_R1_peak_6    452     22.60400    48.2640    45.24710      132
-----
seqinfo: 17 sequences from an unspecified genome; no seqlengths
> resize(peaks[2:6], 1, fix = 'center')
GRanges object with 5 ranges and 6 metadata columns:
      seqnames      ranges strand |      name      score signalValue      pValue      qValue      peak
      <Rle> <IRanges> <Rle> | <character> <numeric> <numeric> <numeric> <numeric> <integer>
[1]   chrI      92683      * | Reb1_R1_peak_2    118      8.90459    14.3647    11.80700      105
[2]   chrII      5969      * | Reb1_R1_peak_3     63      6.16472     8.7750     6.36486       51
[3]   chrII    111307      * | Reb1_R1_peak_4   1714     63.70210   175.6310   171.48900       76
[4]   chrII    124931      * | Reb1_R1_peak_5    397     20.54910    42.7364    39.77400       99
[5]   chrII    135918      * | Reb1_R1_peak_6    452     22.60400    48.2640    45.24710      132
-----
seqinfo: 17 sequences from an unspecified genome; no seqlengths
> reduce(peaks[2:6])
GRanges object with 5 ranges and 0 metadata columns:
      seqnames      ranges strand
      <Rle>      <IRanges> <Rle>
[1]   chrI    92560-92807      *
[2]   chrII     5846-6092      *
[3]   chrII  111226-111389      *
[4]   chrII  124859-125004      *
[5]   chrII  135791-136046      *
-----
seqinfo: 17 sequences from an unspecified genome; no seqlengths
```

Comparison functions

- %over%
- distance()
- distanceToNearest()
- findOverlaps()
- subsetByOverlaps()

```
> distanceToNearest(peaks[1:5], Reb1_hits)
Hits object with 5 hits and 1 metadata column:
```

	queryHits	subjectHits	distance
	<integer>	<integer>	<integer>
[1]	1	1673	0
[2]	2	427	5386
[3]	3	2081	0
[4]	4	14	0
[5]	5	1124	0

queryLength: 5 / subjectLength: 3642

```
> findOverlaps(peaks, Reb1_hits)
Hits object with 446 hits and 0 metadata columns:
```

	queryHits	subjectHits
	<integer>	<integer>
[1]	1	450
[2]	1	895
[3]	1	1620
[4]	1	1673
[5]	3	516
...	...	...
[442]	365	216
[443]	365	1803
[444]	367	288
[445]	368	197
[446]	369	2763

queryLength: 369 / subjectLength: 3642

Comparison functions

- %over%
- distance()
- distanceToNearest()
- findOverlaps()
- subsetByOverlaps()

```
> table(peaks %over% Reb1_hits)

FALSE TRUE
   36   333

> subsetByOverlaps(peaks, Reb1_hits)
GRanges object with 333 ranges and 6 metadata columns:
      seqnames      ranges strand |      name      score signalValue      pValue      qValue      peak
      <Rle>      <IRanges> <Rle> |      <character> <numeric>      <numeric> <numeric> <numeric> <integer>
[1]      I      102-492      * | Reb1_R1_peak_1      81      6.32656      10.6179      8.16077      28
[2]     II     5846-6092      * | Reb1_R1_peak_3      63      6.16472      8.7750      6.36486      51
[3]     II 111226-111389      * | Reb1_R1_peak_4     1714     63.70210     175.6310    171.48900      76
[4]     II 124859-125004      * | Reb1_R1_peak_5      397     20.54910     42.7364     39.77400      99
[5]     II 135791-136046      * | Reb1_R1_peak_6      452     22.60400     48.2640     45.24710     132
...      ...      ...      ... |      ...      ...      ...      ...      ...
[329]    XVI 829041-829232      * | Reb1_R1_peak_364     147     10.27450     17.3633     14.74180     129
[330]    XVI 840491-840698      * | Reb1_R1_peak_365      63      6.13093      8.8075      6.39684      98
[331]    XVI 870371-870586      * | Reb1_R1_peak_367     292     16.43930     32.0800     29.23800     161
[332]    XVI 899847-900090      * | Reb1_R1_peak_368    1279     45.43150    131.8010    127.92500     175
[333]    XVI 942584-942868      * | Reb1_R1_peak_369     162     10.95950     18.9055     16.25210     111
-----
seqinfo: 17 sequences from an unspecified genome; no seqlengths
```

Biostrings in R

```
> seqs <- Biostrings::readDNAStringSet('Share/day03/results/bwa/mergedLibrary/macs/narrowPeak/Reb1_R1_peaks.narrowPeak.fa')
> seqs
DNAStringSet object of length 369:
      width seq
[1] 391 CCAACCTGTCTCTCAACTTACCCTCCATTACCCTGCCTCCACTCGTT...ATATACCATCTCAAACTTACCCTACTCTCAGATTCCACTTCACTCCA I:101-492
[2] 248 TACTGCTAAACTTCGAGATATTTTCGAATTTTTCAGTCTTTTCTTTT...CTAACTGTTACCTTTTGAAATAAAATAAGGGGAAGGTCAAAAAGCTA I:92559-92807
[3] 247 ATACCCTAACACTACCCTAACCTACCCTATTTCAACCCTTCCAACC...TTCACTACCACTTACCCTGCCATTACTCTACCATCCACCATCTGCTA II:5845-6092
[4] 164 TTCATCTCTTTGTAAATAGTGTTATACCATAGTAGTAGTTTCAATAA...GAACGGAAGGGGTTTAATAGTTGTATGCTTAACATATTTGATTAA II:111225-111389
[5] 146 AATCTCAGCTGAAAGGCTGCCTTTAATTGTTATTCTTTTCCAGGAAA...AATCTATTACCTCGGATTAACCTGAATTAATAAGGACACACAGGTAT II:124858-125004
...
[365] 208 ACTTACTGGTCTTTAGCACACGACGACCGTACTTTGCACGTGGCTGC...TTCAGACCCACACAAAATCCGCGTAGCCGAGATTGCTTATGTATGTT XVI:840490-840698
[366] 152 AAGGGGTATGTTCTCAGCATTATCTGAAGGTACTCCTCTAAATTTT...ATAATATCAGGTAAAGAAATTGTTGGAATAAAAATCCACTATCGTCT XVI:844286-844438
[367] 216 AGGAAAAAAGGAAAAAGCAAAAAATATCGATTTTATGACTTACAAA...TACCCGCATATTATCGGGAAACAGAAGCCATGTTAGAGTGATTTCCA XVI:870370-870586
[368] 244 TAGTCGTCGCAAGCGACAAATCTCAACTGACAGTAAATAACGGTGAT...TTCTTGTTCCACCTCTTTTCCCCAACATATATGAACATGAGATGGTA XVI:899846-900090
[369] 285 TGGGTGAATGGCACAGGGTATAGACCGCTGAGGCAAGTGCCGTGCAT...GAAGCGTGAGGTCTGATACCTAATAAGGAAATGTAATTTATACTTT XVI:942583-942868
```



```
> seqs[2:5]
DNAStringSet object of length 4:
      width seq
[1] 248 TACTGCTAAACTTCGAGATATTTTGAATTTTTAGTCCTTTCTTTT...CCTAACTGTTACCTTTTGAAATAAAATAAGGGGAAGGTCAAAAAGCTA I:92559-92807
[2] 247 ATACCTAACAACCTACCCTAACCCTACCCTATTTCAACCCTTCCAACCT...CTTCACTACCACTTACCCTGCCATTACTCTACCATCCACCATCTGCTA II:5845-6092
[3] 164 TTCATCTCTTTGTAAATAGTGTTATACCATAGTAGTGTTCATAAT...AGAACGGAAGGGGTTTAATAGTTGTATGCTTAACATATTTGATTAA II:111225-111389
[4] 146 AATCTCAGCTGAAAGGCTGCCTTTAATTGTTATTCTTTCCAGGAAAA...TAATCTATTACCTCGGATTAACCTGAATTAATAAGGACACACAGGTAT II:124858-125004

> reverse(seqs[2:4])
DNAStringSet object of length 3:
      width seq
[1] 248 ATCGAAAACTGGAAGGGGAATAAAATAAAGTTTTCCATTGTCAATCC...TTTTTCTTTTCTGACTTTTTTAAGCTTTTATAGAGCTTCAAATCGTCAT I:92559-92807
[2] 247 ATCGTCTACCACCTACCATCTCATTACCGTCCCATTACCATCACTTC...TCCAACCTTCCCAACTTTATCCCATCCCAATCCCATCACAATCCCATA II:5845-6092
[3] 164 AATTAGCTTTATACAATTGATGTTGATAATTTGGGAAGGCAAGA...TAATAACTTTGATGATGATACCATATTGTGATAAATGTTTCTCTACTT II:111225-111389

> reverseComplement(seqs[2:4])
DNAStringSet object of length 3:
      width seq
[1] 248 TAGCTTTTTGACCTTCCCCCTTATTTTATTTCAAAGGTAACAGTTAGG...AAAAAGAAAAGACTGAAAAATTGAAAATATCTCGAAGTTTAGCAGTA I:92559-92807
[2] 247 TAGCAGATGGTGGATGGTAGAGTAATGGCAGGGTAAGTGGTAGTGAAG...AGGTTGGAAGGGTTGAAATAGGGTAGGGTTAGGGTAGTGTAGGGTAT II:5845-6092
[3] 164 TTAAATCGAAATATGTTAAGCATACAACCTATTAACCCCTTCCGTTCT...ATTATTGAAACTACTACTATGGTATAACACTATTTACAAAGAGATGAA II:111225-111389

> width(seqs[2:4])
[1] 248 247 164

> names(seqs[2:4])
[1] "I:92559-92807" "II:5845-6092" "II:111225-111389"
```